3. In a two-sector economy (with only households and firms), the consumption function is given by C = 10 + 0.75 Y and $\bar{I} = 20$.

   (i) Derive the saving function.

   (ii) Derive equilibrium output in terms of the equality of planned saving and investment.

   (iii) Suppose there is an increase in the desire to save while planned investment remains unchanged. The new saving function is given by S = – 8 + 0.25 Y. Find out the new equilibrium income level.

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2.5 LIQUIDITY PREFERENCE

To complete our understanding of the Keynesian model we have to discuss the concept of 'liquidity preference' or the demand for money in this system. An important contribution of Keynes was to point out the interest sensitivity of the demand for money. Note that demand for money refers to the demand for 'real balances', i.e., nominal money demand deflated by the general price level. Since prices are constant in the Keynesian system, any change in money demand is essentially a change in the demand for real balances.

2.5.1 Components of Demand for Money

There are three main reasons people wish to hold money. The main components of the demand for money are (i) transactions demand, (ii) precautionary demand, and (iii) speculative demand. Let us discuss each of the above.

Transaction Demand: People hold cash since it is a medium of exchange and is useful for making transactions. This gives rise to transactions demand for money. You would have noticed that people with higher income tend to hold more money for transactions compared to the poor. So it is assumed that the amount of liquid cash people wish to hold for transactions is directly proportional to their incomes.

Precautionary Demand: People wish to hold money as it would be useful in contingencies such as medical emergency, accident, etc. It is reasonable to assume that precautionary demand for money is also an increasing function of income.

Speculative Demand: According to Keynes, people also hold money for speculation in the asset market, which gives rise to speculative demand for money. To understand why speculative demand for money is sensitive to the rate of interest, consider an economy where there are only two types of financial assets – money and bonds. Money is useful for transactions, but it yields no returns, while bonds offer interest income and possibility of making capital gains when bond prices rise. There is an inverse relation between the price of bonds and rate of interest. Now consider the effect of change in interest rate on speculative demand for money. When interest rate is high, people will mostly expect it to fall and expect bond prices to rise. Therefore, they would reduce money holdings and buy bonds in the expectation of making a capital gain. So, at high rates of interest, demand for money would be low. Conversely, at low rates of interest, people expect interest rates to rise (i.e., fall in bond prices). Capital loss would induce people to sell bonds and increase the demand for money. Therefore, speculative demand for money is *inversely* related to the rate of interest.

In general, the higher the market rate of interest, the higher is the opportunity cost (interest income forgone) of holding money. So when interest rates rise, people would reduce money demand and try to hold interest-bearing assets instead which explains the inverse relation between interest rates and the demand for money. This is another way of understanding the inverse relation between money demand and interest rates, without assuming the presence of speculative demand.

From our discussion above, you should see that the money demand function can be expressed as follows:

$$L = q\,Y - t.i \qquad \ldots (2.15)$$

where $q > 0$, $t > 0$; L is the demand for real balances, Y is aggregate income and i is the rate of interest. Diagrammatically the money demand function can be shown as a downward sloping line in Fig. 2.10, where interest rate is presented along the vertical axis and money demand is along the horizontal axis.

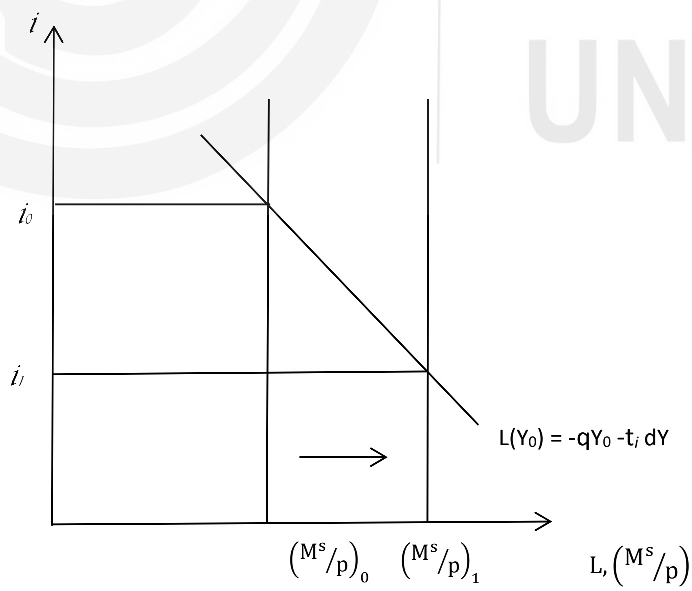

Fig. 2.10: Demand for Money

Note that the money demand function $L(Y_0)$ represents demand for money at the income levelY_0. Movement along $L(Y_0)$ demonstrates the interest sensitivity of the money demand function. As interest rate falls from i_0 to i_1, there is an increase in the demand for money due to speculative demand for money, as discussed above. For all income levels, demand for money greater than Y_0 would be higher at each interest rate (since transactions and precautionary demands for money increase with income), and the money demand curve would shift to the right. For incomes lower than Y_0, money demand would be lower at each interest rate and the money demand curve would shift to the left.

Money supply is determined by the central bank of the country (you will learn more about it in subsequent Units) and is given by the vertical line M^s/P, where M^s is nominal money supply, P is the price level and M^s/P is the real money supply. Equilibrium in the money market occurs at the intersection of money supply and money demand. Note that with the downward sloping money demand function, an increase in money supply (from $(M^s/P)_0$ to$(M^s/P)_1$) brings about a reduction in the rate of interest (from i_0 to i_1).

2.5.2 Liquidity Trap

A special situation can arise if interest rates become unusually low and hit the lowest possible level, so that the demand for money becomes infinitely elastic and the economy faces a 'liquidity trap'. This can happen when interest rates have fallen so low (and bond prices have risen so high) that everyone expects interest rates to rise (and bond prices to fall). Therefore no one wants to hold bonds, in fear of making a capital loss and everyone wants to hold money. In this situation, the money demand curve becomes horizontal as presented in Fig. 2.11 and *t* in equation (2.16) tends to infinity.

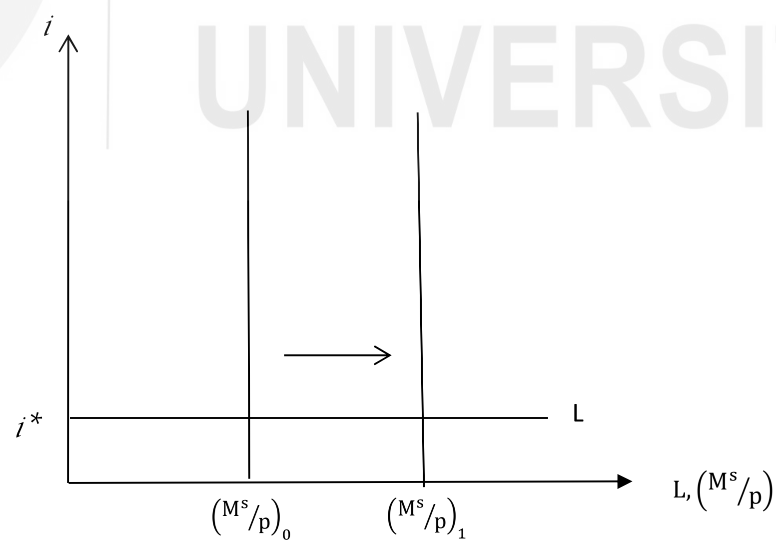

Fig. 2.11: Liquidity Trap